

SSIP

(Sword Suffle Internet Protocol)

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Introduction

The Sword Suffle Internet Protocol (SSIP) version 1.0 is a lightweight, UDP-based protocol designed to synchronize player states and transient game events in real-time multiplayer environments. SSIP ensures low-latency communication between a central game server and multiple clients while maintaining consistency across all participants, even under moderate network loss or delay.

Unlike traditional reliable transport protocols such as TCP, SSIP prioritizes timeliness over guaranteed delivery, making it suitable for fast-paced, real-time applications such as online multiplayer games. The protocol achieves reliability through selective redundancy and ordered snapshot handling rather than retransmission.

Use Case:

SSIP is used in simplified multiplayer games, where multiple players interact with a shared grid world. The server broadcasts frequent game state snapshots, while clients apply interpolation or smoothing techniques to maintain a consistent visual experience despite packet loss or delay.

Assumptions and Constraints:

- Transport layer: UDP
- Maximum packet size: 1200 bytes (header + payload)
- Average end-to-end latency target: ≤ 50 ms
- Supports up to 4 concurrent clients
- Tolerates random packet loss up to 5%
- Clients discard outdated or duplicate snapshots
- Periodic update rate: 10–60 Hz

SSIP 1.0 emphasizes minimal overhead, efficient encoding, and temporal consistency to support smooth multiplayer experiences on local or wide-area networks.

Protocol Architecture

The **Sword Suffle Internet Protocol (SSIP)** uses a client-server model over UDP. The server maintains the main game state and sends periodic snapshots to all clients. Clients send event messages back to the server.

Entities

- Server: Maintains the authoritative state, broadcasts updates, and manages snapshot order.
- Client: Receives and applies snapshots, sends player actions, and smooths state changes to reduce jitter.

Reliability

SSIP applies selective reliability:

- Critical events require acknowledgments.
- Regular snapshots use redundancy to handle packet loss.
- Outdated or duplicate snapshots are discarded.

State Flow

Client:

Idle → Receive Snapshot → Update State → Idle

Server:

Collect Events → Update World → Broadcast Snapshot → Repeat

Message Formats

Each SSIP message contains a fixed binary header followed by an optional payload. The header provides essential information for synchronization, reliability, and ordering.

Field	Size (bytes)	Description
protocol_id	4	4-character ASCII identifier (“LABA”)
version	1	Protocol version number
msg_type	1	Type of message (SNAPSHOT, EVENT, ACK, CONNECT, KEYFRAME, ID_SET)
snapshot_id	4	Snapshot identifier for ordering updates
seq_num	4	Packet sequence number
server_timestamp	8	Server send time in milliseconds
payload_len	2	Length of payload data
checksum	4	CRC32 checksum for error detection

Sample Message:

protocol_id	version	msg_type	snapshot_id	seq_num	server_timestamp	payload_len	checksum
LABA	1	0	15	102	1730900000000	20	0x5A3C9D02